Ref. [47] — Global Optimality in Neural Network Training

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1. Problem: when does a non-convex neural objective have a global certificate?

The training of a neural net is non-convex in its layer parameters, but some overparameterized structures have a convex interpretation after regrouping their components. Haeffele and Vidal ask for *sufficient conditions* ensuring that certain local minima are global, and for a non-increasing path from an arbitrary starting parameterization toward a global solution. This does not claim that all stationary points of arbitrary deep nets are global minima.

2. Factorized parallel networks and positive homogeneity

A representative structured model is a sum of r parallel subnetworks:

$$\Phi_r(W) = \sum_{i=1}^{r} \phi(w_i), \qquad \Theta_r(W) = \sum_{i=1}^{r} \vartheta(w_i).$$

Training takes the schematic form

$$\boxed{\min_W \ \ell\left(Y, \sum_{i=1}^{r} \phi(w_i)\right) + \lambda \sum_{i=1}^{r} \vartheta(w_i).}$$

The mapping ϕ and its size-controlling regularizer ϑ are assumed *positively homogeneous with matching degree*:

$$\phi(cw) = c^p \phi(w), \qquad \vartheta(cw) = c^p \vartheta(w), \qquad c \geq 0.$$

A ReLU neuron satisfies $\mathrm{ReLU}(cz) = c\mathrm{ReLU}(z)$ for $c \geq 0$, but a general network with biases and arbitrary normalization does not automatically satisfy the *joint parameter* homogeneity required here. The degree and the regularizer must match.

3. Why adding a zero component can help

A zero-valued subnetwork can create a direction in the *factorization space* through which infinitesimal parameter changes realize a new functional component. The paper proves global-optimality conditions for certain local minima in sufficiently large homogeneous parallel models when such a component is present; it also discusses a non-increasing path to a global solution. *Geometric intuition, not a full proof:* imagine you have a representation $X = \sum_i \phi(w_i)$ and can split or reallocate the representation across more parallel components without raising the regularized objective. Extra components make a non-convex parameterization flexible enough to connect to an equivalent convex-gauge description. The original proof uses specific atomic/gauge constructions and duality, not merely the assertion that "more neurons improve optimization."

4. Why the matching degree matters: derivation from the supplement

The official supplemental material gives a particularly instructive counterexample when the output map has degree p but the regularizer has smaller degree $p' < p$. For a fixed nonzero direction W and small $s > 0$,

$$\phi(sW) = s^p \phi(W), \qquad \vartheta(sW) = s^{p'} \vartheta(W).$$

For convex ℓ with a subgradient at the origin, a first-order lower bound gives the *form*

$$\ell(s^p \phi(W)) + s^{p'} \vartheta(W) - \ell(0) \ \geq \ s^p \langle q, \phi(W) \rangle + s^{p'} \vartheta(W).$$

As $s \downarrow 0$, the positive regularization term of order $s^{p'}$ dominates the potentially negative loss change of order s^p. Thus the origin can become a *local minimum* even when it is not globally best. This demonstrates why the theory cannot be transferred to a deep ReLU model with an arbitrary degree-mismatched weight penalty.

Main result. Under structured parallel architecture, convex loss in the output, appropriate homogeneity/regularization, and sufficient model size, special local minima (notably those containing an inactive subnetwork) admit a global-optimality certificate. The paper also analyzes paths toward global solutions.

5. What was and was not verified

The full CVPR paper and its *official eight-page proof supplement* are available. The supplement is directly relevant to the homogeneity restrictions; its Proposition 2 supplies the mismatch counterexample summarized above. This note explains the central theorem mechanism and its limits, but it is not a theorem-by-theorem proof check of every proposition. Do not replace the specified model class with "any sufficiently wide nonlinear neural net."

6. Survey relevance and limitations

Use in **RQ1** (sufficient conditions for benign minima) and **RQ4** (nontrivial assumptions needed to transfer such results to realistic models). General feedforward networks with unstructured depth, biases, arbitrary regularizers, or insufficient width are not covered by this claim. Nor does the result imply every generic SGD trajectory efficiently finds an optimal solution in wall-clock time.

> **Scope.** The result does *not* say all local minima are global for general ReLU/Transformer training. The global statement is conditional and depends on the network's parallel factorization and homogeneous regularizer.

7. Our notes

The useful idea here is that extra homogeneous subnetworks can make a non-convex parameterization behave like a convex problem, but only when the architecture and regularization fit together precisely. The proof supplement even shows why using the wrong degree of regularization can create a spurious minimum at zero. That's a strong example for our theory-versus-practice discussion. **Paper:** https://openaccess.thecvf.com/content_cvpr_2017/html/Haeffele_Global_Optimality_in_CVPR_2017_paper.html.
Supplement: https://openaccess.thecvf.com/content_cvpr_2017/supplemental/Haeffele_Global_Optimality_in_2017_CVPR_supplemental.pdf.